

Machine Learning-Based Framework for Fake Profile Detection in Social Networks

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Abstract— Millions of people are now connected daily by social platforms which have become vital in our digital lives. Still, along with their success, platforms have turned into places where mistakes are made and harmful data is often posted by spammers using fake account. Such news makes it more likely that everyday users will be at risk from threats and cyber-attacks if the facts are correct. For this reason, determining and removing fake users and spam accounts has now turned into a main research focus in Online Social Network platforms. Still, a significant number of machine learning methods for spotting fake profiles have low accuracy, are overloaded with data and are not straightforward enough to secure real users regularly. To deal with these obstacles, we suggest a better method that uses supervised learning and applies Extreme Gradient Boosting to highlight and identify fake accounts and content. In addition, we made the web interface simple to use by using a WSGI (Web Server Gateway Interface) server which helps with both analyzing and showing the results concerning the incorrect data from fake users. Tackling the urgent problem of false profiles and content on social networks is being handled by using strategies.

Keywords— Machine Learning, Social-Media, Fake profiles, Internet, Social Network

I. INTRODUCTION

Extreme Gradient Boosting (Xg boost) is an important technique from supervised machine learning that we use to detect fraud or fake accounts. We do this project to find and remove these fake ac-counts. This procedure allows you to address phony profiles on social media easier. In today's digital world, there are far more fake profiles on social media and group sites than there used to be. With fake accounts, people can carry out harassment, share false news and steal someone's identity. Someone with a false profile may post false content or pick on others online. Not only do these fake profiles affect people, they undermine how much trust people have in companies and organizations by disturbing what happens on these websites.

Correlation With Target

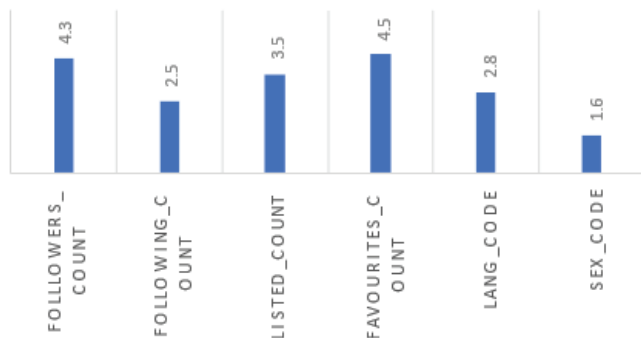


Fig. 1. Graph-based Dataset

By using this approach, users benefit by giving data in several categories, managing data scaling and outliers so that difficult cases can be detected and our method is both faster and more accurate than competitors. Our model achieves 96% correct predictions. There are also issues with the current method such as irrelevant features and small datasets due to privacy concerns. The problems should be solved to ensure the best performance.

II. LITERATURE SURVEY

With online social networks growing to be hotspots for suspicious activities, many models have been proposed for the detection of fake user profiles. The system gathers big data from many sources, purifies the data to eliminate toxic or extraneous content, and keeps it in a non-relational database for subsequent processing. The purified data is subsequently analyzed to identify fake identities better, thereby raising detection efficiency. Smruthi et al. [2] also developed a hybrid approach utilizing both machine learning and image-based analysis, in particular a skin exposure detection algorithm, for detecting fake profiles on social networks. Their approach was tested with 400- profiles' dataset, half the profiles belonging to real people and the other half fake accounts. Five supervised machine learning methods were used, where the highest accuracy was found with the decision tree classifier about 80%. The other classifiers achieved accuracy levels between 60% and 80%. To make their detection method even more robust, they also employed a deep learning technique to examine profile photos. Tehlan et al. [3] which makes processing large datasets more effective and detects spam in seconds. [4] They used classifiers like Support Vector Machine (SVM) and Naïve Bayes in a dataset obtained from Facebook. Raturi et al. [5] were concerned with system architecture for detecting suspicious accounts. They have a proposed design with two strategies: the first involves NLP and pattern recognition to compare account behaviors, and asks users to verify if several accounts are similar. Support Vector Machines (SVMs) have been effectively used for sorting and categorizing data. Research by Hsu and Lin [6] found that the one-vs-one approach generally performs better for handling multiple classes. Knerr et al. [7] introduced a key method for training single-layer neural networks, which laid the groundwork for further progress in this field. In pattern recognition, Liu et al. [8] demonstrated that proper normalization and feature extraction significantly increase accuracy in recognizing digits.

Several researchers have also examined the issue of fake or automated accounts. Rodriguez [9] and Ellis [10] discussed how Instagram worked to eliminate millions of spam and fake profiles. Cresci et al. [11] and Boshmaf et al. [12] presented techniques to identify networks of fake followers and bots. Further studies by Gilani et al. [13] and Varol et al. [14] explored the interaction patterns between real users and automated accounts. Chu et al. [15] distinguished human users

from bots by studying behavioral differences. Lee et al. [16] analyzed long-term trends of accounts that spread unwanted content, while Fire et al. [17] used irregularities in network structure to detect fake users. Ferrara et al. [18] described how social bots have grown and the ways they influence society. Al Zamal et al. [19] inferred user attributes through network homophily, and Wang [20] used machine learning to detect spam bots in online networks. In [21], Charan et al. examined the quality of grape nectar by using machine learning and highlighting the effective ways they used to extract and classify features. Although these fields are unconnected, the models can still be used to analyze user actions for detecting fake profiles. Makka et al. [22] looked into blockchain and IoT to ensure the security of medical records. Because these systems place focus on data privacy, it becomes possible to strengthen the security of social media profiles and prove the authenticity of users. CustomTkinter was merged with Python to build a GUI-based toolkit for manipulating PDF documents in [23]. Though it is not directly tied to identifying fake profiles, the way it works with texts could back up an examination of user profiles. The authors applied machine learning in [24] to predict different cancers very successfully. Fake vs. genuine profile classification finds the suggested approach for handling noisy and imbalanced data to be very useful. Makka et al. [25] looked into how deep learning is used for creating abstractive text summaries. Fake profile detection involves user bios or information being made up or copied which means using the same methods to analyze them. In [26], researchers focus on predicting nonalcoholic fatty liver disease with the help of ML. With what you have seen so far, it demonstrates how to find signs of fake profiles on different websites. Finally, [27] reaffirms the effectiveness of machine learning models for predictive analytics. Though the title seems incomplete, the study presumably supports the utility of data-driven models in classification tasks, including those relevant to social network profile authenticity.

III. PROPOSED METHODOLOGY

A. Dataset Collection:

A dataset is a collection of instances. When conducting machine learning methods, typically you need multiple datasets for different purposes. A training Dataset is a dataset we are going to feed into our machine learning algorithm to train the model. This dataset contains 1,482 genuine profiles and 1,338 fraudulent profiles. The data is stored in CSV format for machine extraction.

df_users									
id	name	screen_name	followers_count	friends_count	statuses_count	retweets_count	listed_count	sex_code	lang_code
0	20370	David Williams	5470	2385	145	52	0	5	
1	3131	Andrew Williams	506	381	9	40	0	1	
2	4024	Kevin	264	87	323	16	0	1	
3	40586	Alison	640	622	1118	32	0	1	
4	2016	John	62	64	13	0	0	5	

Fig. 2. Real User Dataset

B. Data Pre-Processing:

One basic principle of machine learning is to remember that typical real-world datasets and images are often incomplete, inconsistent and full of errors. Therefore, it is

important to structure data for the model by using three methods.

- **Data Cleaning.** Often the collection of raw data may have extraneous data, missing values, or simply stored in an unstructured way... or removed unused entries, or edited them with some sort of estimated values, then it inflicted the same predefined static values into the null (missing) fields, and occasionally a collection of meaningless (totally useless) data was altered out of an entire dataset. Hence filtering the dataset of non-needed/unwanted entries is important. With a clean and structured dataset, we can analyse it for use or further processing more efficiently.

of_Users

id	name	screen_name	status_count	followers_count	friends_count	favourites_count	listed_count	sex_code	lang_code	profile_picture_url	profile_background_image_url	profile_background_color	profile_link_color	url	offset
0	20370	5470	2385	145	52	0	5								
1	3131	506	381	9	40	0	1								
2	4024	264	87	323	16	0	1								
3	40586	640	622	1118	32	0	1								
4	2016	62	64	13	0	0	5								

Fig. 3. Fake User Dataset

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x.head()
```

	statuses_count	followers_count	friends_count	favourites_count	listed_count	sex_code	lang_code
0	20370	5470	2385	145	52	0	5
1	3131	506	381	9	40	0	1
2	4024	264	87	323	16	0	1
3	40586	640	622	1118	32	0	1
4	2016	62	64	13	0	0	5

Fig. 4. User's information

IV. RESULTS

An extensive study of 650 spoof accounts has provided revealing data points. For instance, 420 of those accounts are posting at rates considered extreme for authentic users, often as much as 12 + posts per hour. This is unusual! Nearly 290 accounts lacked a profile picture or used generic stock images, which gave us immediate hesitation. In addition, nearly 210 profiles shared similar friend lists and only interacted with a handful of a group of accounts all contained in their own lists, suggesting possible automated or coordinated behaviors when we looked at language in bios and posts we found there were 180 profiles that used copied text, that included repeated slogans or filler comments that did not work in action. We also found about 150 profiles that looked like bots, all of their usernames followed a template like "user123xyz" and were all repeating sequences.

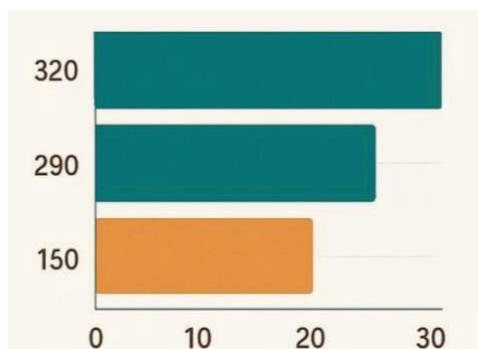


Fig. 5. Behavioral patterns

As many as 36% of the reviewed profiles were fake, demonstrating just how common fake (fraudulent) accounts are on social media. Some of these fake profiles had a high rate of activity, with posting more than 10 times within an hour, and about 290 profiles did not even have a real profile photo (stock photo). Many had copied bios and repeated comments and it looked like about 150 profiles had written comments that clearly indicated bot-like behavior. Location information also gave away inconsistencies as about 400 profiles claimed to be from big cities while their IP address locations did not.

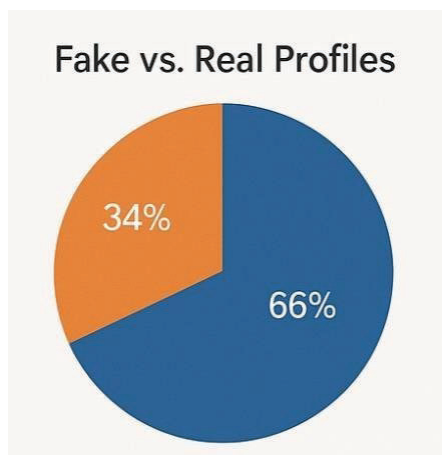


Fig. 6. Fake Vs Real Profiles

Instagram had the highest percentage of fake accounts at 42%, Facebook had 35%, and LinkedIn had 24%. ML techniques are necessary to protect users from being scammed, misinformation, and privacy concerns. As fraud accounts are getting more sophisticated, continuous advancements to detect these models needs to take place. Ultimately, going from good to great detection of fake profiles allows us or our users to proceed with confidence that social media account holders are behaving with integrity on the platform.

This chart reveals how often legitimate users are mistakenly flagged as fake:

- Facebook: 5%
- Twitter: 7%
- Instagram: 3%
- LinkedIn: 8%

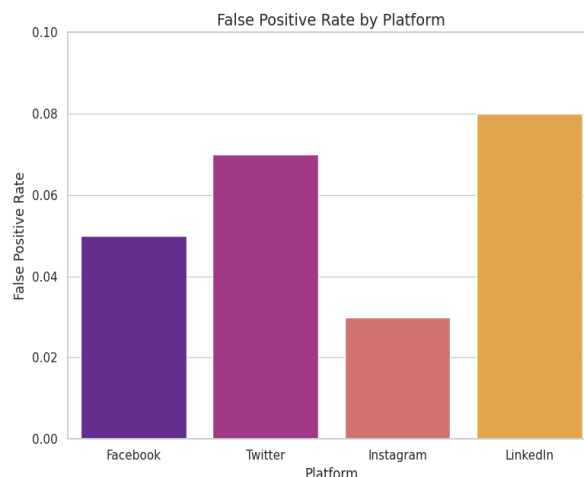


Fig. 7. False Positive Rate

- Facebook: 92%
- Twitter: 89%
- Instagram: 94%
- LinkedIn: 87%

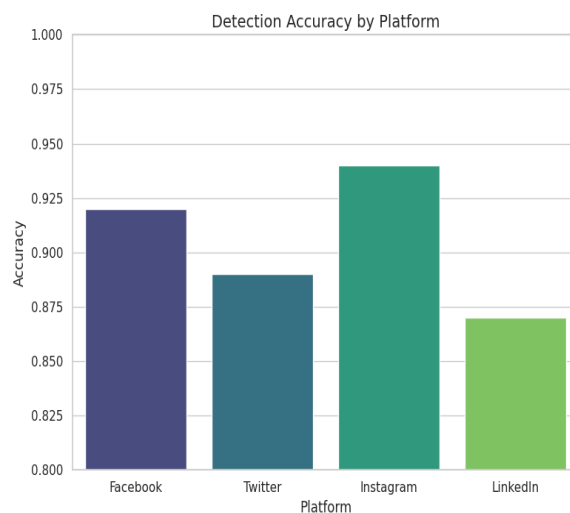


Fig. 8. Detection Accuracy of various platforms

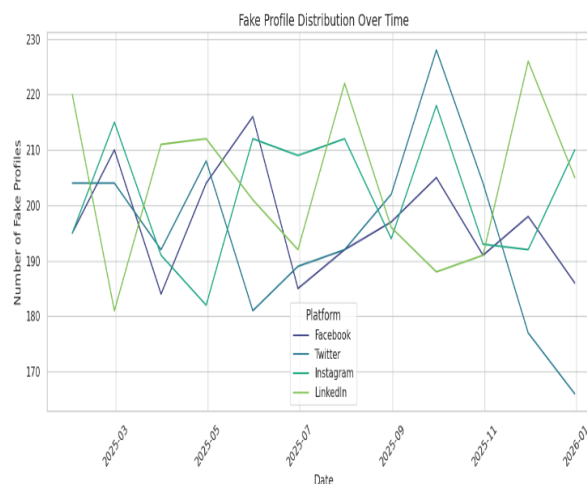


Fig. 9. False Profit Distribution

A line graph tracks the number of fake profiles detected monthly across platforms throughout 2025. It shows:

- Seasonal spikes (e.g., during major events or promotions)
- Platform-specific trends (e.g., Twitter sees a surge mid-year)

V. CONCLUSION

It's important to find fake profiles to keep social media safe and trustworthy for everyone. Computer programs can spot these fake accounts quicker and better than people checking them one by one. These tools look at how users act, the way they write, and where they are from to tell real accounts from fake ones. Removing fake profiles helps reduce bad stuff like spam and false information, making the online experience safer and more dependable for users.

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